

Large Language Model Fundamentals

Understand what an LLM predicts, how text becomes tokens, and why generation is uncertain.



WHY THIS MATTERS

Start from what you already know

Python, vectors, probability, neural-network layers, and inference

- 01 Explain tokens and context windows in plain language.
- 02 Use next-token prediction in a small worked example.
- 03 Complete the guided lab and verify base and instruction models.

CORE CONCEPT

The vocabulary of this topic

Understand what an LLM predicts, how text becomes tokens, and why generation is uncertain.

tokens and context windows: Tokens are model vocabulary units; the context window limits how many input and output tokens can be considered.

next-token prediction: An LLM assigns probabilities to possible next tokens from the tokens already in context.

logits and softmax: Logits are raw token scores; softmax converts them into positive probabilities that sum to one.

base and instruction models: A base model predicts text continuations; an instruction model is adapted to follow conversational tasks.

FOUR CONNECTED IDEAS

How the concept works

01

**tokens and context
windows**



02

**next-token
prediction**



03

**logits and
softmax**



04

**base and
instruction models**

$P(\text{token}_i | \text{previous tokens}) = \text{softmax}(\text{logits})$; generation repeatedly samples or selects a next token.

CORE CONCEPT

Read every part of the mechanism

$P(\text{token}_i | \text{previous tokens}) = \text{softmax}(\text{logits})$; generation repeatedly samples or selects a next token.

token: A vocabulary unit represented by an integer.

logit: An unnormalized model score.

temperature: A control on sampling sharpness.

WORKED EXAMPLE

Worked example by hand

For logits $[2, 1, 0]$, subtract 2, exponentiate, and normalize to approximately $[0.665, 0.245, 0.090]$.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
import numpy as np
logits = np.array([2.0, 1.0, 0.0])
p = np.exp(logits - logits.max())
print(np.round(p / p.sum(), 3))
```

OUTPUT

```
[0.665 0.245 0.09 ]
```

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model

- ☒ Text becomes token IDs, then vectors; the model predicts a distribution for the next token.

- ☒ Softmax probabilities come from relative logits and always sum to one.

- ☒ Temperature changes sampling sharpness but does not add knowledge or factual verification.

- ☒ Base models continue text; instruction models are adapted to follow task-oriented conversations.

FOUR CONNECTED IDEAS

Notebook readiness map

01

**tokens and context
windows**



02

**next-token
prediction**



03

**logits and
softmax**



04

**base and
instruction models**

Use the concepts from this lesson to read and complete
<https://github.com/curiously/Al-Bootcamp/blob/master/05.llms-101.ipynb>.

RISK REVIEW

Common beginner mistakes

- 01 Treating generated text as retrieved knowledge.
- 02 Changing temperature without understanding that it changes sampling, not factual knowledge.

GUIDED LAB

Tokenize a prompt, inspect next-token probabilities, and compare deterministic and sampled local generation.

01

02

03

04

Predict

Run

Inspect

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

- 01 Explain tokens and context windows in plain language.
- 02 Use next-token prediction in a small worked example.
- 03 Complete the guided lab and verify base and instruction models.

NEXT EVIDENCE

A local Ollama notebook documenting tokens, context, generation settings, outputs, and observed limitations.